Clinical Findings and Diagnosis: Cutaneous Manifestations of Estrogen Excess and Deficiency

Estrogen exerts both beneficial and adverse cutaneous effects. In postmenopausal women, estrogen replacement therapy can reverse epidermal atrophy and restore cutaneous collagen. Estrogen's inhibitory effect on sebum production contributes to the improvement of acne seen with oral contraceptive therapy, particularly formulations with higher estrogenic activity. However, despite elevated estriol levels during pregnancy, the relative androgen excess—driven in part by progesterone's ability to bind androgen receptors—can exacerbate acne in some individuals.

Telogen effluvium, characterized by diffuse shedding of scalp or body hair, may occur postpartum or following discontinuation of oral contraceptives. Both pregnant women and those taking synthetic estrogens are more susceptible to estrogen-related side effects, including telangiectasias, palmar erythema, spider angiomas, and pigmentary changes.

Melanocytic lesions such as preexisting nevi and malignant melanoma may darken during pregnancy. Hyperpigmentation commonly affects the nipples, linea nigra, and genitalia. **Melasma**, often termed the "mask of pregnancy," presents as irregular, hyperpigmented patches on the forehead, cheeks, nose, upper lip, chin, and neck. Although melasma may improve after delivery, it often recurs in subsequent pregnancies. When induced by oral contraceptives, melasma can persist even after stopping the medication.

Acanthosis nigricans may be associated with estrogen excess. In conditions with elevated estrogen levels—such as pregnancy or oral contraceptive use—increased hair growth on the face, breasts, and extremities may occur. This is likely due to the complex interaction between estrogens and androgenic hormones rather than a direct stimulatory effect of estrogen on hair follicles.

Estrogen and progesterone dermatitis is a rare condition characterized by a polymorphous, cyclical eruption that flares pre-menstrually and may include pruritus. Diagnosis can be supported by intradermal testing with estrogen and progesterone, with observation for erythema or wheal formation within 15 to 60 minutes. Oral contraceptives have also been linked to **erythema nodosum**.

Estrogen deficiency—due to menopause, pituitary failure, or medications such as danazol, leuprolide, or clomiphene citrate—can lead to hot flashes, genital epithelial atrophy, and decreased breast size. Flushing episodes, marked by intense heat lasting several minutes, may also occur in younger women during the late luteal phase when estrogen levels are lowest.

Treatment

The Pigmentary Disorders Academy recommends fixed "triple-ingredient" combinations as first-line therapy for melasma. Topical treatments commonly used include retinoids, corticosteroids, azelaic acid, and hydroquinone-based bleaching agents.

Kligman's formula, developed in 1975, is the classic triple-combination preparation containing: - 5% hydroquinone (a depigmenting agent), - 0.1% tretinoin (a retinoid that enhances epidermal turnover), - 0.1% dexamethasone (a corticosteroid that reduces inflammation).

This formulation is typically compounded in a hydrophilic ointment base. Dual-ingredient combinations or single-agent therapies are acceptable alternatives when triple therapy is not tolerated or contraindicated.